Dries Van De Putte

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Skills Profile

- Observational astronomy: Writing, reviewing, and implementing HST and JWST programs. Use of pipelines for imaging and IFU data, including NIRCam, NIRSpec IFU, MIRI imager, MIRI IFU. Analysis of emission lines and broader molecular emission features in NIR and MIR spectra and IFU data cubes. Use of UV spectroscopy from HST STIS and IUE.
- Development and use of reproducible and well-documented spectral analysis tools; interacting with users of those tools. Decomposition of spectra into continua and emission features. Bayesian SED fitting to determine stellar properties and dust extinction.
- Development of astrophysical simulations: 3D dust radiative transfer using Monte Carlo techniques. Implementing recipes for the interactions between radiation, dust, and gas in the interstellar medium. Solving for thermal balance, chemical networks, level populations.
- Computer skills: Development of Python packages, git and GitHub, continuous integration, unit testing. Creating interactive Jupyter notebooks demonstrating the use of analysis tools. Use of data visualization tools (matplotlib, GlueViz). Linux command line, HPC clusters, multi-processing with MPI, simulation development in C/C++.
- Communication skills: Collaborative development (open-source), working in large and small international collaborations, speaking at conferences and seminars.

Research Experience

2024—* **Postdoctoral Associate** University of Western Ontario, Canada

- Currently leading work on JWST observations of the C60 molecule in the central cavity of NGC 7023.
- Led PDR-GTO paper IV (under review), concerning the decomposition and spatial distribution of carbonaceous emission bands in the NGC 7023 photodissociation region.
- Led PDR4All paper XVI (2025), documenting improvements made to the PAHFIT spectral decomposition tool, its application to JWST spectra of the Orion Bar, and the use of emission band profile diagnostics as tracers of photochemical evolution.
- Took a leading role in the PDRs4All data reduction team and set up an automated packaging script and versioning system for updated data products.
- PI for the the approved HST observing program "Linking Gas-Phase Element Depletions and Extinction Curves in the Small Magellanic Cloud" (GO-17823).
- Created a Python package with tools shared by our research group. Performed maintenance tasks including automated testing, documentation, and performing code review.
- Advised researchers concerning the data reduction of JWST spectra and the application of PAHFIT in several research projects. Mentored graduate students assisting with Python programming and JWST data analysis.

2021—2024 **Postdoctoral Researcher** Space Telescope Science Institute, Baltimore, MD

- Contributed to JWST spectroscopic data reduction in two large international collaborations ("PDRs4All" JWST-ERS-1288 and "PDR-GTO" JWST-GTO-1192). Set up the PDRs4All Python package, containing scripts to run the JWST pipeline and several postprocessing steps. The data reduction team frequently contacted the JWST helpdesk to report and resolve issues with early version of the pipeline. Identified issues and developed workarounds for the pipeline, and delivered high-level data products to research collaborators to support numerous research papers.
- Led PDRs4All paper V (2024), a high-impact (23 citations), collaboration-wide (100 members) overview paper of the mid-IR line inventory.
- Major contributor and contact person for the open-source spectral decomposition tool PAHFIT. Led the redesign of the user-facing API, contributed to work making PAHFIT compatible with JWST data, and developed a demo notebook serving as a user guide. Served as a contact person to help users apply PAHFIT to their research.

- Led a research project analyzing UV spectroscopy data from space telescopes (HST, FUSE, IUE) to reveal the link between far-UV optical properties of interstellar dust and H_2 in the Milky Way (Van De Putte et al. 2023).

2016—2020 **Ph.D. in Astronomy** Ghent University, Belgium
Space Telescope Science Institute, Baltimore, MD

- Four year fellowship obtained by submitting a research plan and application to BOF (*bijzonder onderzoeksfonds*), the "special research fund" provided by Ghent University.
- *Ph.D. thesis:* "Self-consistent modelling of radiation, dust, and gas in the interstellar medium". Developed RADAGAST, wrote around 10 000 lines of C++ to model the interactions of interstellar gas with the radiation field and dust grains, using linear algebra (`Eigen`) and differential equation (GSL) solvers. Coupled RADAGAST to the open-source C++ code SKIRT, to enable 3D radiative transfer simulations that include both interstellar gas and dust, in the context of photodissociation region modeling. Supervision: Maarten Baes, Karl Gordon, Julia Roman-Duval.
- Led a research project at STScI in collaboration with the ISM*@ST group, where I applied a Bayesian stellar and dust model (BEAST) to HST photometry data, revealing dust evolution in the IC 63 nebula. Contributed to the open-source BEAST code by implementing a new model parameter (stellar distance) and domain splitting tools to reduce the memory impact.
- Teaching assistant for the course "Statistics and data processing" for two semesters (Fall 2018, 2019).

Education

2011—2016 **B.Sc. & M.Sc. in Physics & Astronomy** Ghent University, Belgium

- *Master's thesis:* "Data parallelization with MPI in a 3D Monte Carlo dust radiative transfer code". Set up distributed-data multi-processing in the SKIRT C++ code, splitting up the computational domain to reduce memory usage across compute nodes. This enabled simulations with that would normally exceed the memory capacity of a single node. Implemented efficient inter-process communications using the industry-standard Message Passing Interface (MPI), to switch between memory distribution schemes as required for the algorithm. Co-authored a research article by Verstocken et al. (2017) by including a description of this method. *Supervision:* Maarten Baes.
- *Internship at the Royal Observatory of Belgium:* Developed a new pipeline step in Python for the HERMES spectrograph of the Mercator Telescope, to remove biases introduced by the spectral properties of the flat-field calibration lamp.
- *Notable elective courses:* Software Development I: C and C++; Simulations and modeling on the nano-scale.

Teaching and Mentoring

- Led tutorial sessions and mentored graduate students for our research group at Western University, mainly concerning Python programming, JWST data reduction practices, and spectroscopic analysis.
- Fall 2018 and 2019: Teaching assistant for a second year Bachelor course of Ghent University "Statistics and data processing". Supervised the exercise sessions, and set up three Python programming challenges which introduced the students to essential packages such as *numpy* and *scipy*, and numerical statistical tasks such as error propagation and bootstrapping.

Awards

Observing time

- PI of Hubble Space Telescope program GO-17823 (55 orbits); "Linking Gas-Phase Element Depletions and Extinction Curves in the Small Magellanic Cloud".
- Co-I for a set of successful JWST and HST proposals including:
  GO-8007 (Tielens et al.): Characterizing C60 cations in the cavity of NGC 7023
  GO-7772 (Peeters et al.): Interstellar PAHs: separating the neutrals from the ions

GO-7396 (Tarantino et al.): Exploring the Last Watering Hole of Low Metallicity PAH Emission: Deep Spectroscopic Observations of Sextans A
GO-5168 (Gordon et al.): First Direct Measurements of Interstellar Silicate Dust in the Magellanic Clouds
GO-5952 (Roman-Duval et al.): Zooming-in on the sub-grid physics of PAHs at 20% solar metallicity
GO-4551 (Gordon et al.): MIR Extinction for ice sightlines with measured UV extinction
GTO-1192 (Misselt et al.): Physics and Chemistry of PDR Fronts
GO-17516 (Decleir et al.): Dust extinction at its extremes - from the smallest to the largest dust grains
GO-17488 (Yanchulova Merica-Jones et al.) Unraveling the mystery of the rare 2175 A extinction bump in the SMC

Funding

- 2024: US$157k budget approved for HST GO-17823 (U.S. admin PI: K. Gordon).
- 2016: Four-year Ph.D. fellowship: Ghent University Bijzonder Onderzoeksfonds ("exceptional research fund").

Service

- 2025—2026: Referee for three papers in Astronomy & Astrophysics, one paper in The Astrophysical Journal.
- JAN 2025: External panel member for JWST cycle 4 time allocation committee.

Selected Publications

1. Van De Putte, Gordon, Misselt, et al. in review — *JWST observations of photodissociation regions. IV. Carbonaceous emission band sub-components in NGC 7023 have distinct spatial distributions*
2. Van De Putte, D., Peeters, E., Gordon, K. D., et al. 2025, A&A, 701, A111 — *PDRs4All. XVI. Tracing aromatic infrared band characteristics in photodissociation region spectra with PAHFIT in the JWST era*
3. Van De Putte, D., Meshaka, R., Trahin, B., et al. 2024, A&A, 687, A86 — *PDRs4All. VIII. Mid-infrared emission line inventory of the Orion Bar*
4. Van De Putte, D., Cartledge, S. I. B., Gordon, K. D., Clayton, G. C., & Roman-Duval, J. 2023, ApJ, 944, 33 — *Far-ultraviolet Dust Extinction and Molecular Hydrogen in the Diffuse Milky Way Interstellar Medium*
5. Van De Putte, D., Gordon, K. D., Roman-Duval, J., et al. 2020, ApJ, 888, 22 — *Evidence of Dust Grain Evolution from Extinction Mapping in the IC 63 Photodissociation Region*
6. Verstocken, S., Van De Putte, D., Camps, P., & Baes, M. 2017, Astronomy and Computing, 20, 16 — *SKIRT: Hybrid parallelization of radiative transfer simulations*

Talks and Conferences

2025/12/04 — London, ON	**Invited seminar**	*Western University physics and astronomy colloquium*
2025/09/25 — Kingston, ON	**Invited seminar**	*Queen's University physics and astronomy seminar*
2025/11/10 — Tucson, AZ	Poster	*Dusty Universe 2025* "Carbonaceous emission band sub-components in NGC 7023 have distinct spatial distributions"
2025/06/12 — Baltimore, MD	Poster	*Inter+Stellar: Harnessing the Intersection Between Stars and the Interstellar Medium* "Spectral Decomposition of the Aromatic Infrared Bands in Photodissociation Regions"
2023/11/14 — Baltimore, MD	Poster	*Improving JWST Data Products Workshop:* "Tools and corrections for imaging and IFU spectra of the Orion Bar"
2023/10/29 — Florence, Italy	**Contributed talk**	*Illuminating the Dusty Universe: A Tribute to the Work of Bruce Draine:* "Spatially resolved maps of the IR emission bands in photodissociation regions"
2023/09/26 — Gothenburg, Sweden	**Contributed talk**	*Origin and Fate of Dust in Our Universe:* "Far-UV extinction and H_2; Decomposing and mapping IR bands near the HI-H_2 transition"

2022/12/16 — Remote	Webinar	*JWebbinar 23: PDRs4ALL Community Telecons in Support of JWST Cycle 2 Proposals; Webinar 3: Tools To Quantify the Spectral Information:* Live PAHFIT demo
2022/11/30 — Ghent, Belgium	**Contributed talk**	*SKIRT Days 2022:* implementation of the RADAGAST gas model and its use in SKIRT
2022/07/11 — Paris-Saclay, France	Discussion	*Interstellar Institute #5 - With Two Eyes*
2022/06/12 — Pasadena, CA	Poster	*240th Meeting of the American Astronomical Society:* Poster about the far-UV exinction rise to H_2 relationship
2022/03/15 — Baltimore, MD	Seminar	*STScI Discovery Seminar Series:* "Far-UV Dust Extinction and Molecular Hydrogen in the diffuse Milky Way Interstellar Medium"
2019/06/24 — Lyon, France	Poster	*EWASS2019, Cosmic dust (r)evolution:* IC 63 project
2018/11/05 — Leiden, Netherlands	Poster	*Hendrik van de Hulst Centennial Symposium:* IC 63 project
2018/06/11 — Copenhagen, Denmark	Poster	*CPHDUST2018: Cosmic Dust: origin, applications & implications:* IC 63 project
2016/10/28 — Florence, Italy	**Contributed talk**	*DustPedia meeting:* the parallelization of SKIRT
2016/10/04 — Ghent, Belgium	**Contributed talk**	*CHARM meeting:* the parallelization of SKIRT